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MILITARY SPECIFICATION

ALTERNATOR-RECTIFIER

(For Torpedo Mk 46 Mods 0 and 1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification covers one type of brushless alternator-rectifier, referred to herein as the "alternator."

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

LD 546973	Alternator-Rectifier, and all documents listed thereon (Torpedo Mk 46 Mod 0)
LD 269801	Alternator-Rectifier, and all documents listed thereon (Torpedo Mk 46 Mod 1)

FSC 1355

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LD 269676	Alternator-Rectifier, Shipping (Packing) Condition and all documents listed thereon (Torpedo Mk 46 Mod 0)
DL 1330516	Alternator-Rectifier, Shipping (Packing) Condition and all documents listed thereon (Torpedo Mk 46 Mod 1)
LM 1330509	Alternator-Rectifier and all documents listed thereon (Torpedo Mk 46 Mod 1)
Dwg 2574648	Regulator, Alternator, Voltage

PUBLICATIONS

NAVAL ORDNANCE SYSTEMS COMMAND

QAP 6024	Quality Assurance Provisions for the Alternator-Rectifier

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other documents. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

PUBLICATIONS

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, the preproduction sample shall:

(a) Be manufactured using the procedures proposed for production.

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(b) Consist of two alternators.

(c) Be tested as specified in Section 4 herein and is for the purpose of determining, prior to starting production, that the supplier's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 Basic requirements

3.2.1 Contractor-furnished data. Data required by this specification, or by applicable documents referenced in Section 2, need not be furnished unless specified in the contract or order (see 6.2).

3.2.2 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the alternator shall be in accordance with the requirements specified herein and with the applicable documents listed in LD 546973, LD 269801, or LM 1330509, as applicable.

3.2.3 Gages. The supplier shall provide whatever gages are necessary and adequate to ensure that the material and parts of the alternator will meet dimensions and requirements shown on the applicable documents listed in LD 546973, LD 269801, or LM 1330509, as applicable. When requested by the supplier, the procuring activity will furnish existing pertinent drawings of Naval final inspection gages for use only in the designing of the supplier's gages. This action does not relieve the supplier of the responsibility for designing and manufacturing the gages needed for satisfactory fulfillment of the contract requirements.

3.2.4 Inspection and test equipment and facilities. The supplier shall furnish and maintain all necessary inspection and test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate to permit performance of the required quality conformance tests.

3.3 Performance requirements and physical characteristics

3.3.1 Performance. The alternator shall meet the performance requirements specified herein while operating at the indicated speeds and with the shunt field windings energized by an external power supply or a voltage regulator, Drawing 2574648, as specified.

3.3.1.1 Alternating current output. The alternator shall provide 32 plus or minus 4 volts alternating current (vac) root-mean-square (rms) with a frequency of 2000 cycles per second (cps) across a resistance of 1000 plus or minus 50 ohms at a rotor speed of 24,000 revolutions per minute (rpm) while being excited and loaded as in 3.3.1.2.1.

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3.3.1.2 Direct current output using an external power supply

3.3.1.2.1 Steady state load. The alternator, with a rotor speed of 20,000 rpm and the field excited with a current of 1.05 plus 0.20 or minus 0.35 amperes by an external power supply, shall develop a potential of 40 plus or minus 0.3 volts direct current (vdc) average, from minus to common, across a resistive load of 10 plus or minus 0.5 ohms after a maximum of three pulses (see 3.3.1.2.2). Simultaneously, the potential from plus to common shall be within plus or minus 1.0 vdc of the absolute value of the voltage, from minus to common, across a 10 plus or minus 0.5 ohm resistance.

3.3.1.2.2 Pulse loading. The alternator, after conforming to 3.3.1.2.1, shall respond to an additional pulsed load in accordance with the tolerances of Figure 1c. Both plus and minus voltages of the steady state and pulse circuits are to be measured at the reference points of Figure 2. The pulse load resistance shall be 1.81 plus or minus 0.01 ohms, applied to the pulse circuit as a step function for a duration of 60 plus 10 minus 5 milliseconds (ms).

3.3.1.3 Direct current output using a voltage regulator

3.3.1.3.1 Steady state loads. The alternator, with the shunt field excited by the voltage regulator (3.3.1), shall develop the following potentials across the specified load resistances and at the specified rpm when measured at the reference points shown on Figure 3.

(a) Load resistance. 12 plus or minus 0.6 ohms resistance from the plus steady state line to common; 10 plus or minus 0.5 ohm resistance from the minus steady state line to common.

(b) 16,000 rpm. At 16,000 revolutions per minute (rpm) the potential shall be 40 plus 1.0 minus 1.5 vdc across the 10 ohm resistance. The potential across the 12 ohm resistance shall be within 2 volts of that across the 10 ohm resistance.

(c) 20,000 rpm. At 20,000 rpm through 25,200 rpm the potential shall be 40 plus or minus 0.5 vdc across the 10 ohm resistance. The potential across the 12 ohm resistance shall be within plus or minus 1 volt of that across the 10 ohm resistance.

(d) Ripple voltage. The ripple voltage, measured from the steady state lines to common (see Figure 3), shall not exceed 2 volts peak-to-peak.

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3.3.1.3.2 Pulse loading. The alternator, while conforming to 3.3.1.3.1(c), shall respond to an additional pulsed load of 3200 watts minimum in accordance with Figures 1a and 1b when measured at reference points shown on Figure 3. The pulse load shall be applied to the pulse circuit at 0.5 plus or minus 0.1 second intervals for 30 pulses minimum before evaluation of performance. The ripple voltage, measured from the pulse lines to common, shall not exceed 1.0 volt peak-to-peak between pulses.

3.3.1.4 Overspeed. The alternator shall be capable of a 27,000 rpm speed for 5 minutes without abnormal vibration or damage while developing the following potentials: The potential shall be 40 plus 1.0 minus 1.5 vdc across the 10 ohm resistance. The potential across the 12 ohm resistance shall be within plus or minus 2.0 volts of that across the 10 ohm resistance.

3.3.1.5 Acceleration. The alternator shall be capable of accelerating from 0 to 90 percent of 24,000 rpm within 1.0 plus or minus 0.2 seconds when loaded as specified in 3.3.1.3.1.

3.3.2 Coolant. The alternator shall be cooled with sea water under the following conditions:

(a) Flow: 1.5 plus or minus 0.3 gallons per minute (gpm).

(b) Temperature: 30 to 38 degrees Fahrenheit (F).

(c) Operating pressure: 75 plus or minus 10 pounds per square inch, gage (psig).

(d) Static proof pressure: 200 plus or minus 5 psig.

(e) Pressure drop through cooling system: 10 pounds per square inch (psi) max.

3.3.3 Insulation resistance. The insulation resistance of all alternator windings and other electrical components shall be 100 megohms minimum to ground (case) and to other circuits, as measured by a current-limiting 50 vdc megohmmeter.

3.3.4 Residual magnetism. The alternator shall have sufficient residual magnetism to produce a potential of 1.5 vdc across the plus steady state line to common when measured 5 minutes after any operation and with the rotor turning at a speed at 16,000 rpm.

3.3.5 Shunt field coil resistance. The resistance of the shunt field coil measured from pins N and R of P5510 shall be 10.0 plus or minus 0.8 ohms.

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3.4 Environmental requirements. Unless otherwise specified, the alternator shall be capable of conforming to the performance requirements and physical characteristics of this specification subsequent to or during each environment, and under all unit conditions (see 6.3.2) specified in this section.

3.4.1 Vibration

3.4.1.1 Vibration, nonoperating. The packed and packaged alternator (see 5.1) shall be capable of withstanding the vibration shown in Table I.

3.4.1.2 Vibration, ready nonoperating and ready. The alternator, mounted as shown in Figure 4 shall be capable of withstanding the vibrations specified in Table I within the temperature range of minus 45 to plus 140 degrees F.

TABLE I

Vibration

Condition	Double Amplitude (In. ±10%) Minimum	Acceleration (g's ±10%) Minimum	Frequency (cps)
Nonoperating	---	1.0	5 - 60
Ready Nonoperating and Ready	0.2 --- 0.036 ---	--- 3.0 --- 10.0	5 - 17 17 - 40 40 - 74 74 - 500

3.4.2 Shock

3.4.2.1 Shock, nonoperating. The packed and packaged alternator shall be capable of withstanding the shock of being dropped on a nonyielding surface from a height of 36 inches.

3.4.2.2 Shock, ready nonoperating and ready. The alternator, mounted as shown in Figure 4, shall be capable of withstanding the shocks specified in Table II.

3.4.3 Acceleration (see 6.3.4). The alternator shall be capable of withstanding the accelerations specified in Table III.

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TABLE II

Shock Schedule

Unit Condition	Peak Magnitude (g's) ±10%	Duration (Milliseconds)	Acceleration Direction*	Number Of Impacts
Ready nonoperating	60	8.0 to 10.0	3 fwd or aft axis 3 left or right axis 3 up or down axis	9
Ready	45 85	18.0 to 20.0 0.5 to 1.5	Long fwd axis 3 up or down axis 3 right or left axis	3 6
	300	0.5 to 1.5	Long aft axis	3

*See Figure 4 for axes of alternator

TABLE III

Acceleration Schedule

Unit Condition	Magnitude (g's) ±10%	Acceleration Direction
Ready nonoperating	35	Longitudinal Aft
Ready	25	Longitudinal Fwd

3.4.4 Operating environment. The alternator shall be capable of operating in an environment of a sea water vapor having a 250 degree F temperature and a 750 psig pressure.

3.4.5 Temperature

3.4.5.1 Temperature, nonoperating and ready nonoperating. The alternator shall be capable of conforming to the requirements of this specification after being at any temperature within the range of minus 45 to plus 140 degrees F.

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3.4.5.2 Ready. The alternator shall be capable of conforming to the requirements of this specification after being at any temperature between 0 and plus 140 degrees F.

3.4.5.3 Operating temperature range. The alternator shall be capable of meeting the performance requirements specified herein while being subjected to an environment in which the ambient temperature rises from 0 to 250 degrees F in the first 2 minutes of operation and stabilizes at 250 degrees F for the remainder of a 10 minute run, using sea water coolant as specified in 3.3.2.

3.5 Life

3.5.1 Service life. The alternator shall have a minimum service life of 7 hours under room ambient conditions. This shall consist of 42 duty cycles as shown in Figure 5. The first 2 minutes of each duty cycle shall be without water cooling and the remaining 8 minutes shall be with cooling in accordance with 3.3.2. Operation during any of the functional tests contained in this document may, with the permission of the procuring activity, be charged to the operational service life of the alternator.

3.5.2 Storage life, ready nonoperating and ready. After storage of one year minimum in the ready nonoperating or ready condition, the alternator, when connected to a voltage regulator, shall deliver the rated output (3.3.1.3), by self-excitation, within 0.5 second after reaching 16,000 rpm.

3.5.3 Storage life, nonoperating. The alternator shall meet the performance requirements specified herein after storage of five years minimum in the nonoperating condition and after the reestablishment of the required level of residual magnetism.

3.6 Identification. The alternator shall be identified in accordance with MIL-STD-130 and the applicable documents listed in LD 546973, LD 269801, or LM 1330509, as applicable.

3.7 Workmanship. The workmanship and finish shall be of sufficiently high grade to ensure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the contract relative to workmanship insofar as not specifically covered by the applicable specifications.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the alternator shall be classified as follows:

(a) Preproduction inspection

(1) Preproduction sample (see 4.3.1)

(2) Preproduction tests (see 4.5)

(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

(c) Quality conformance inspection

(1) Quality conformance sampling (see 4.3.3)

(2) Quality conformance tests and examinations (see QAP 6024)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of two alternators, manufactured in accordance with 3.1, shall be subjected to the preproduction tests detailed in 4.5 at a facility designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the alternator by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. Periodic production samples shall be selected from units having satisfactorily passed the quality conformance tests and examinations of QAP 6024. These samples shall be subjected

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to tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

(a) Step 1. Upon initiation of production randomly select one unit from each group of 30 units. Upon satisfactory compliance of samples selected from three consecutive groups, shift to Step 2. Each sample so selected shall be numbered consecutively starting with Sample Number 1.

(b) Step 2. Randomly select one unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively starting with a sample number that is one greater than the highest sample number selected in Step 1.

4.3.2.1 Defective units. When a defective unit (see 6.3.5) is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(a) Determine the cause of the defect.

(b) Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

(c) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

(d) Submit to the procuring activity for approval the proposed re-test program to verify that the corrective action eliminated the defect.

(e) Return to Step 1 and proceed as indicated when inspection is resumed.

4.3.3 Quality conformance sampling. The tests and examinations detailed in QAP 6024 shall be conducted on each alternator to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the alternator shall be subjected to tests under the following conditions:

(a) Temperature: Room ambient (60 to 95 degrees F).

(b) Humidity (relative): Room ambient, up to 95 percent.

(c) Coolant (clean tap water): Temperature 60 plus or minus 15 degrees F, flow rate 1.5 plus or minus 0.3 gpm.

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4.5 Preproduction tests. The alternators within the preproduction sample (identified as alternator numbers 1 and 2) shall be subjected to the preproduction tests as specified in Table IV. Certified inspection data covering the results of preproduction tests shall be forwarded to the Naval Ordnance Systems Command. Certified inspection data shall consist of the following:

(a) Date of tests and names of persons conducting tests.

(b) Run or test plan used.

(c) Serial number of the alternator.

(d) Test results of preproduction tests (4.5.2 through 4.5.5).

TABLE IV

Preproduction Test Schedule

Test Number	Test Description	Preproduction Sample Number 1	2
1	Quality conformance tests and examinations (QAP 6024)	x	x
2	Minimum rpm operation (16,000) (4.5.2)	x	
3	Overspeed (4.5.3)		x
4	Operational temperature test (4.5.4.1)	x	
5	Temperature, storage (4.5.4.2)		x
6	Acceleration (4.5.4.3)		x
7	Vibration (4.5.4.4		x
8	Shock (4.5.4.5)		x
9	Quality conformance tests and examinations (QAP 6024, except paragraph 20.6.1)		x
10	Operational service life (4.5.5)	x	

4.5.1 Preproduction test equipment. The following items of test equipment are required to perform the preproduction tests specified herein.

(a) All test equipment listed in Section 10 of QAP 6024, with the following exceptions:

(1) The alternator drive gearing must be functionally equivalent to the gear drive provisions on the pertinent Mod 0 or Mod 1 torpedo configurations.

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(2) A separate drive may be used in conjunction with the load and control provisions of the alternator test set in performance of the operational environmental tests. If used, this separate drive must be capable of performance equivalent to that of the alternator test set.

(b) A vibration test fixture capable of subjecting the alternator to vibration as specified in 3.4.1.

(c) A vibration table capable of vibrating a 10 lb load, plus support fixture, at 10 g's maximum over a frequency range of 5 to 500 cps with a maximum double amplitude of 1.0 inch and with a temperature chamber capable of maintaining the test specimen at any temperature between minus 45 plus or minus 5 degrees F and plus 140 plus or minus 5 degrees F.

(d) An acceleration test fixture capable of subjecting the alternator to acceleration as specified in 3.4.3.

(e) A temperature controlled test chamber capable of conditioning the alternator to temperatures from 0 plus or minus 5 degrees F to 250 plus or minus 10 degrees F and subjecting the alternator to the temperature cycle of 4.5.4.2.

(f) A coolant supply system capable of supplying coolant as specified in 3.3.2.

4.5.2 Minimum rpm operation (16,000). Install the alternator on the test equipment with the shunt field connected to the voltage regulator specified in 3.3.1 and a steady state resistive load of 10 plus or minus 0.5 ohms across the negative line of the circuit and 12 plus or minus 0.6 ohms across the positive line of the circuit. Slowly increase the rpm of the rotor until the minus line voltage is minus 40 plus 1.0 minus 1.5 vdc. A rotor rpm of more than 16,000, or a plus line voltage not within plus or minus 2 volts of the absolute value of the negative voltage, constitutes a failure.

4.5.3 Overspeed. Install the alternator on the test equipment. Connect the steady state load of 3.3.1.3.1(a). Accelerate the alternator to 90 percent of 24,000 rpm within 1.0 plus or minus 0.2 seconds, then increase the alternator rpm to 27,000 and maintain that rate for 5 minutes. Output at the end of 5 minutes not conforming to the requirements of 3.3.1.1 and 3.3.1.4, or evidence of damage to the gears, bearings, or housing, constitutes a failure.

4.5.4 Environmental tests. The alternator shall be subjected to the following environmental tests (see Table IV). Unless otherwise specified, inability of the alternator to pass the quality conformance tests

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and examinations, subsequent to the environmental tests, constitutes a failure.

4.5.4.1 Operational temperature test. Mount the alternator in a temperature controlled chamber (see 4.5.1(e)). Conduct six 10-minute duty cycle tests as shown in Figure 5. For the first three tests, condition the alternator at 0 plus or minus 5 degrees F for 2 hours prior to each test. Use sea water coolant at 30 plus or minus 2 degrees F and start each test with an ambient temperature of 0 degrees F. Increase the ambient temperature to 250 degrees F within the first 2 minutes of operation. For the next three tests, condition the alternator at 140 plus or minus 5 degrees F for 2 hours prior to each test, use sea water coolant at 98 plus or minus 2 degrees F, and start each test with an ambient temperature of 140 degrees F. Increase the ambient temperature to 250 degrees F within the first 2 minutes of operation. An output, outside the limits of the performance requirements specified herein, constitutes a failure.

4.5.4.2 Temperature, storage. The alternator, preserved and packaged in accordance with 5.2.1, shall be placed in a temperature controlled chamber (4.5.1(e)) and subjected to the following temperature cycle:

(a) Cool to minus 45 plus 0 minus 5 degrees F within 2 hours.

(b) Maintain at minus 45 plus 0 minus 5 degrees F for 4 hours.

(c) Heat to 140 plus 5 minus 0 degrees F within 2 hours.

(d) Maintain at 140 plus 5 minus 0 degrees F for 4 hours.

(e) Return to room temperature within 2 hours.

Repeat this schedule for a minimum of three cycles.

4.5.4.3 Acceleration. The alternator shall be mounted as shown in Figure 4 and subjected to the acceleration magnitudes and directions specified in Table III for one minute in each direction.

4.5.4.4 Vibration

4.5.4.4.1 Nonoperating. The packaged alternator shall be attached to a rigid support fixture oriented with respect to the three orthogonal axes of the shipping container, one axis being along the cylindrical axis of the container. The resonance of the packaged alternator in each of the three axes shall be determined by varying the frequency of the applied vibration slowly through the specified range of Table I. The packaged alternator shall then be vibrated at the most severe resonance for one hour in each of the three orthogonal axes of the container.

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4.5.4.4.2 Ready nonoperating and ready. The unpackaged alternator, mounted on a support frame, see Figure 4, shall be rigidly attached to the vibration equipment. This support frame shall be capable of transmitting the vibrations listed in Table I to the alternator, along the axis shown in Figure 4. The alternator shall be subjected to 180 minutes of cycling vibration in each of the three axes. Sixty minutes of this vibration in each axis shall be conducted with the alternator conditioned to a temperature of minus 45 plus or minus 5 degrees F, 60 minutes with the alternator conditioned to plus 140 plus or minus 5 degrees F, and 60 minutes with the alternator at room ambient temperature. During cycling, a logarithmic frequency sweep rate shall be used. The sweep rate shall be such that 15 minutes actual vibration time is required to proceed from the lower to higher and back to the lower extremities of the frequency range in accordance with the acceleration or double amplitudes defined in Table I. The level of the input vibration shall be the level of the fundamental input frequency, and a tracking filter shall be used to control this input.

4.5.4.5 Shock

4.5.4.5.1 Nonoperating. The packed and packaged alternator (see 5.1) shall be subjected to free fall drop tests on corners and surfaces of the container. The packaged unit shall be dropped from a height of 36 inches above an impact surface which shall be essentially nonyielding stone, concrete, or similar material. The container, with unit inside, shall be dropped in sequence on corners 5-1-2, 6-3-4, 5-2-3, and 6-1-4, and side 1 and face 5, as defined in Figure 6.

4.5.4.5.2 Ready nonoperating and ready. Mount the alternator onto a shock test fixture, see Figure 4, so that a minimum of 90 percent of table input of the fixture will be applied to the alternator under test. The duration shall be measured by an accelerometer mounted on the fixture to indicate applied shock.

4.5.5 Operational service life. Install the alternator on the test equipment. Conduct 42 duty cycles of 10 minutes each in accordance with 3., and Figure 5. Allow a 10-minute, minimum, cooling period between cycles with the rotor stopped but the coolant flow maintained in accordance with 3.3.2. An output not within the limits specified herein, for any duty cycle, constitutes a failure.

4.6 Periodic production tests. Periodic production samples, selected in accordance with 4.3.2, shall be subjected to tests of 4.5.

4.6.1 Periodic production test equipment. The items of test equipment shall be those specified for all tests of 4.5 and QAP 6024.

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4.6.2 Periodic production test schedule. The periodic production test sample shall be subjected to the tests noted in Table V in the order specified.

4.6.3 Periodic production acceptance criteria. Acceptance criteria shall be in accordance with 4.3.2.1 and QAP 6024, per Table V.

TABLE V

Periodic Production Test Schedule

Test	Sample and Test Sequence		
	Sample No. 1, 4, etc.	Sample No. 2, 5, etc.	Sample No. 3, 6, etc.
Vibration (4.5.4.4)	1*		1*
Acceleration (4.5.4.3)	2*		
Shock (4.5.4.5)	3*	1*	
Temperature, storage (4.5.4.2)	4		2
Hi-Temperature, operate (4.5.4.1)	5**		3**
Lo-Temperature, operate (4.5.4.1)	8**	2**	
Minimum rpm operation (4.5.2)	6		4
Overspeed (4.5.3) (except that acceleration to 24,000 rpm is to be within 1.0 ±0.2 seconds)	7	3	
Operation service life (4.5.5)	9		
Quality conformance tests (QAP 6024, except 20.6.1)	10		5
Quality conformance tests (QAP 6024)		4	

* Indicates visual examinations after test.
** Indicates quality conformance tests following environmental test.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging of the alternator shall be Level A or C, as specified (see 6.2).

5.1.1 Level A. Cleaning, drying, preservation, and packaging methods for Level A shipment shall be in accordance with the instructions contained in the applicable documents listed in LD 269676 or DL 1330516, as applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the supplier's commercial practice, when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level A or C, as specified (see 6.2).

5.2.1 Level A. The alternator, preserved and packaged in accordance with 5.1.1, shall be packed in a container in accordance with the instructions contained in LD 269676 or DL 1330516, as applicable.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The alternator, packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The alternator is intended to be used as a source of direct current power and single phase alternating current at specific voltages and load characteristics.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Required level of preservation and packaging.

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(c) Required level of packing.

(d) Serialization requirements.

(e) Assigned activity for preproduction and periodic production inspection.

(f) That all additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any alternator(s) to pass the prescribed periodic production inspections.

(g) Any exception to this specification.

(h) That certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the Naval Ordnance Systems Command when testing is accomplished at other than a Government activity.

6.3 Definitions

6.3.1 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak. The acceleration is given in multiples of gravitational acceleration (g's). The frequency is given in cycles per second (cps).

6.3.2 Alternator condition

6.3.2.1 Nonoperating. The condition of the uninstalled alternator when stored, transported, or handled.

6.3.2.2 Ready, nonoperating. The condition of the alternator when assembled with the end item, transported, and handled.

6.3.2.3 Ready. The condition of the alternator fully assembled with the end item and installed on the launching mechanism prior to operating.

6.3.2.4 Operating. Operating is the condition of the alternator when operating.

6.3.3 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration (g's), duration in units of time (milliseconds (ms)), and a shock pulse approximating a one-half sine wave. The time duration shall be measured between two points where two lines, tangent at maximum slope, intersect the zero g line.

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6.3.4 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second. Acceleration is given in multiples of gravitational acceleration (g's).

6.3.5 Defective unit. A unit shall be considered defective if it does not meet all examinations or tests specified, or deviations are found through visual examination after environmental exposure.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

6.4.1 Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

6.4.2 Level B. Not applicable.

6.4.3 Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

Preparing Activity
Navy - OS
Project No. 1355-N145

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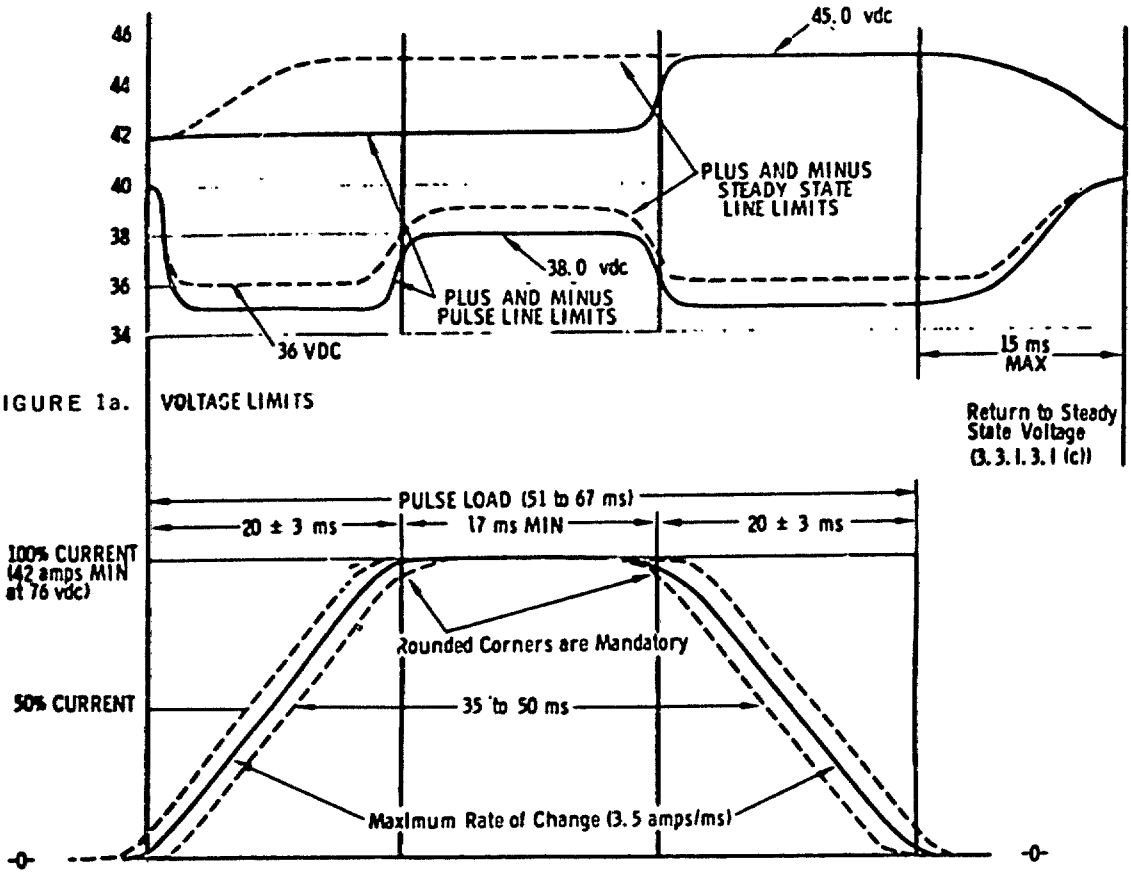


FIGURE 1a. VOLTAGE LIMITS

FIGURE 1b. PULSE CURRENT ENVELOPE

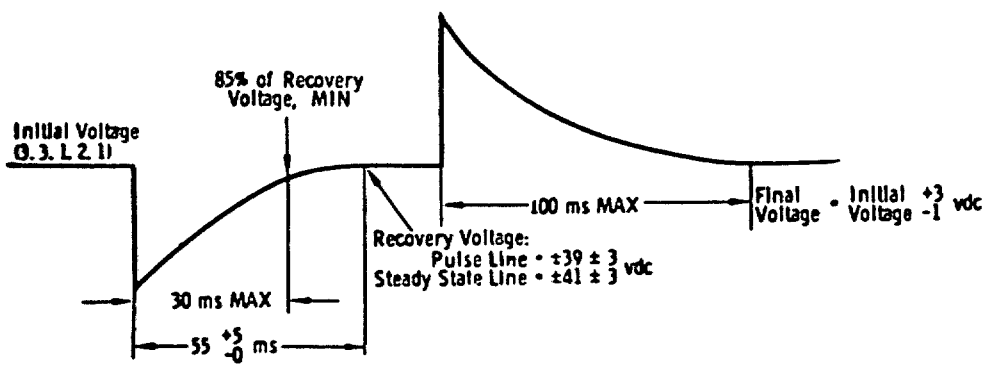
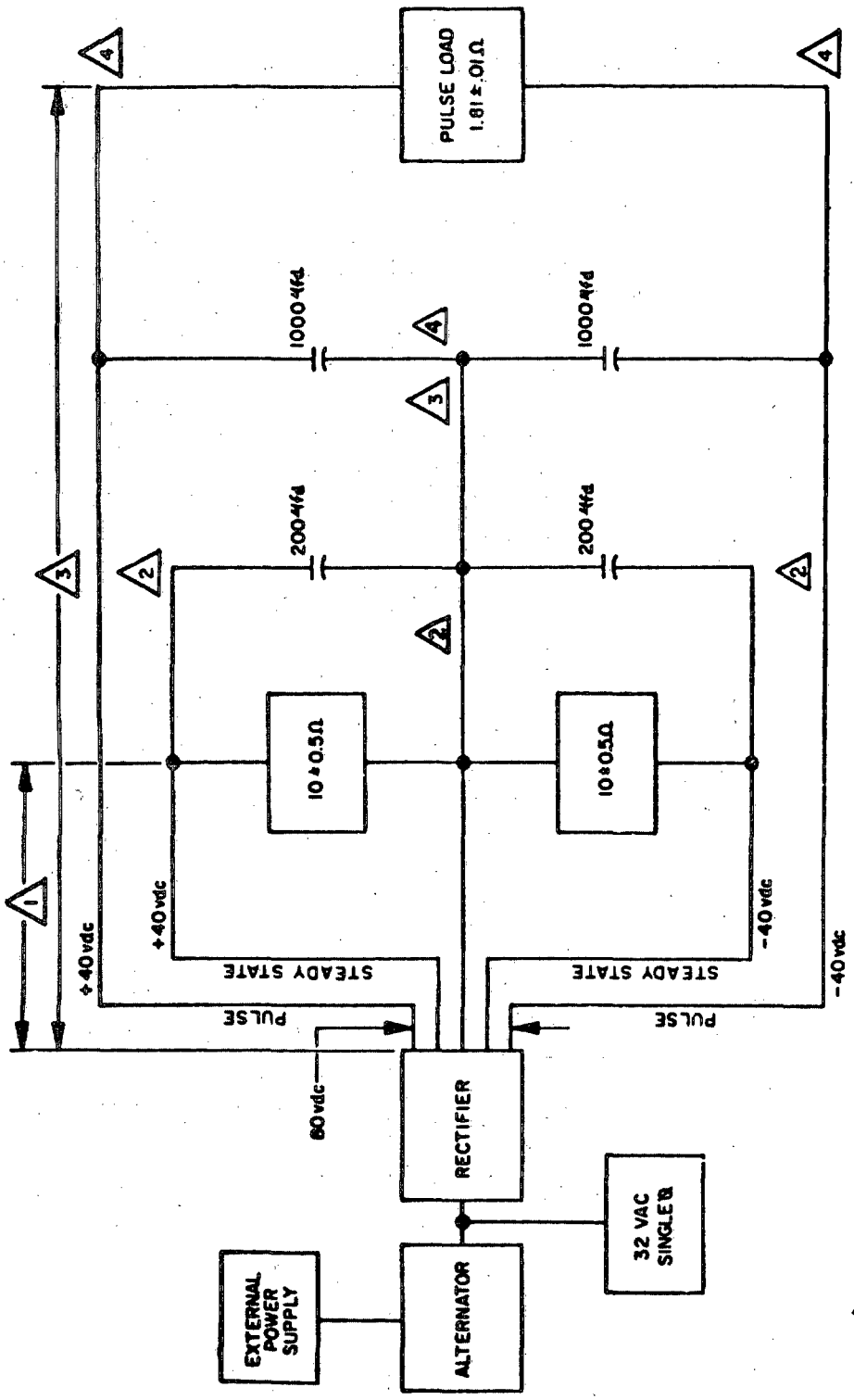


FIGURE 1c. LOAD RESPONSE - UNREGULATED

FIGURE 1. TRANSIENT LIMITS

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1. WIRE RESISTANCE FROM RECTIFIER TO LOAD, APPROX. 0.1 OHM
2. STEADY STATE LINE REFERENCE POINTS
3. APPROX. RESISTANCE OF +40 vdc PULSE LINE 0.02 OHM; -40 vdc PULSE LINE 0.02 OHM, AND COMMON 0.07 OHM
4. PULSE LOAD VOLTAGE REFERENCE POINTS

FIGURE 2. ALTERNATOR-RECTIFIER OUTPUT POWER WITH POWER SUPPLY EXCITATION

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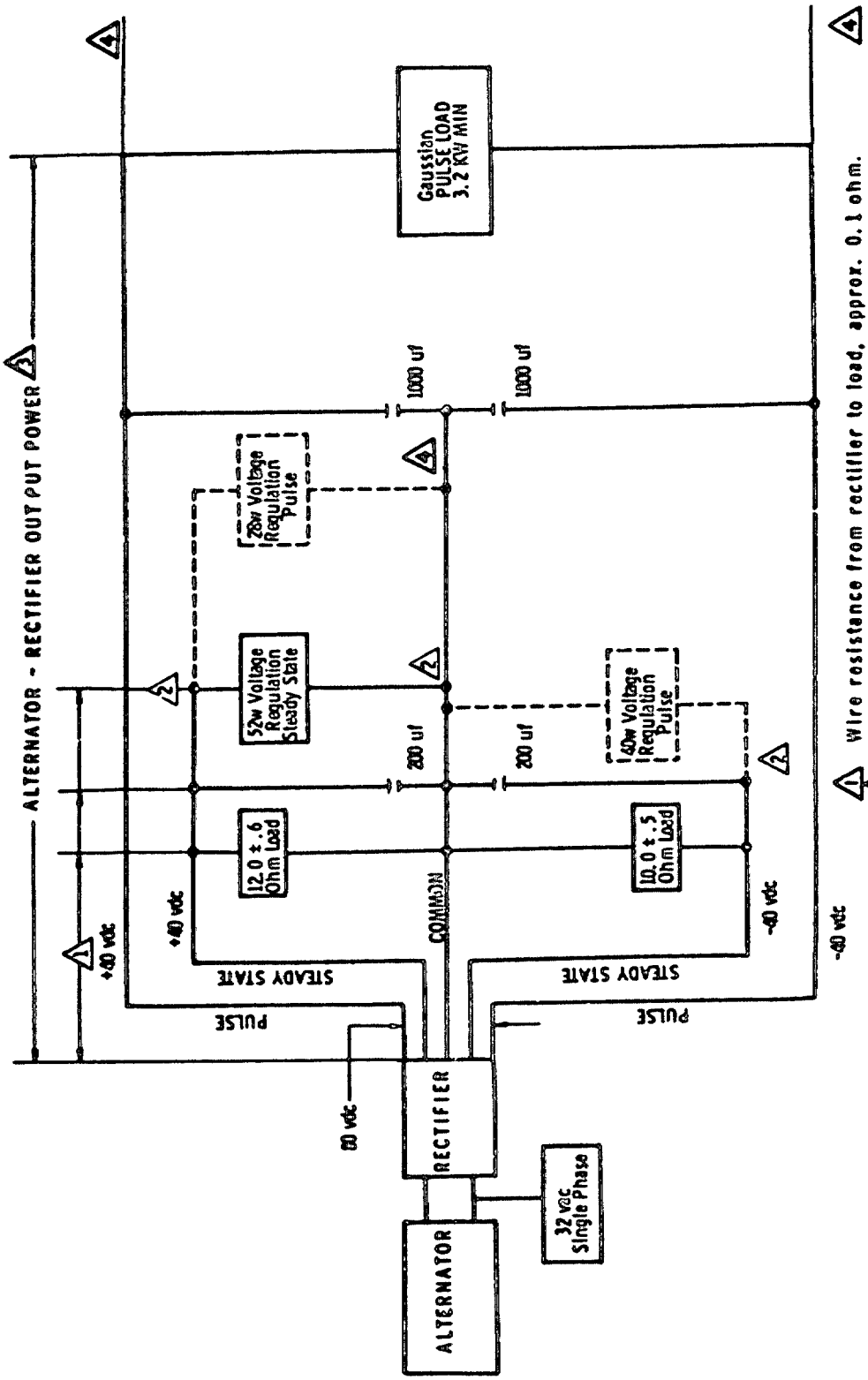


FIGURE 3. ALTERNATOR-RECTIFIER OUTPUT POWER WITH VOLTAGE REGULATOR EXCITATION

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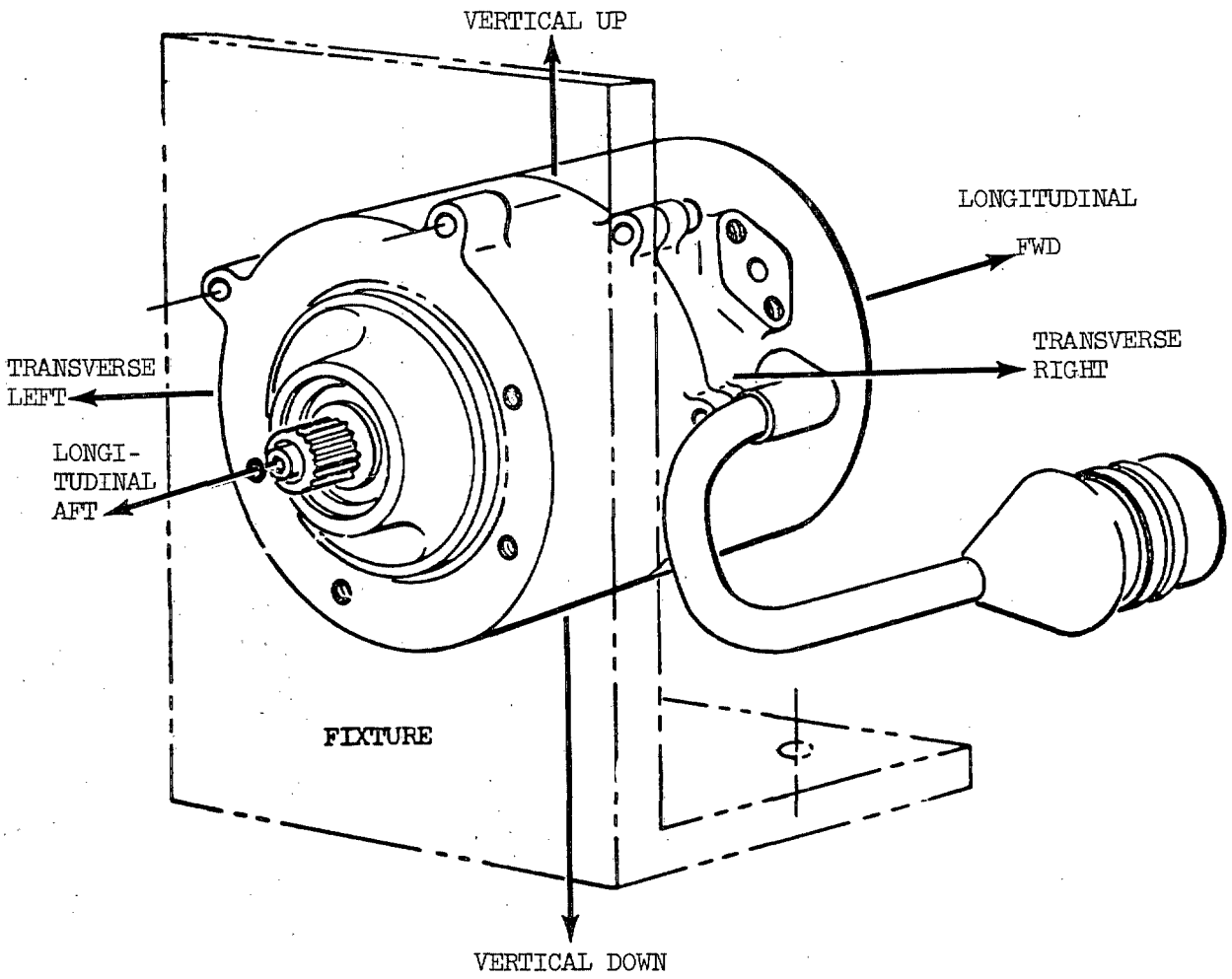
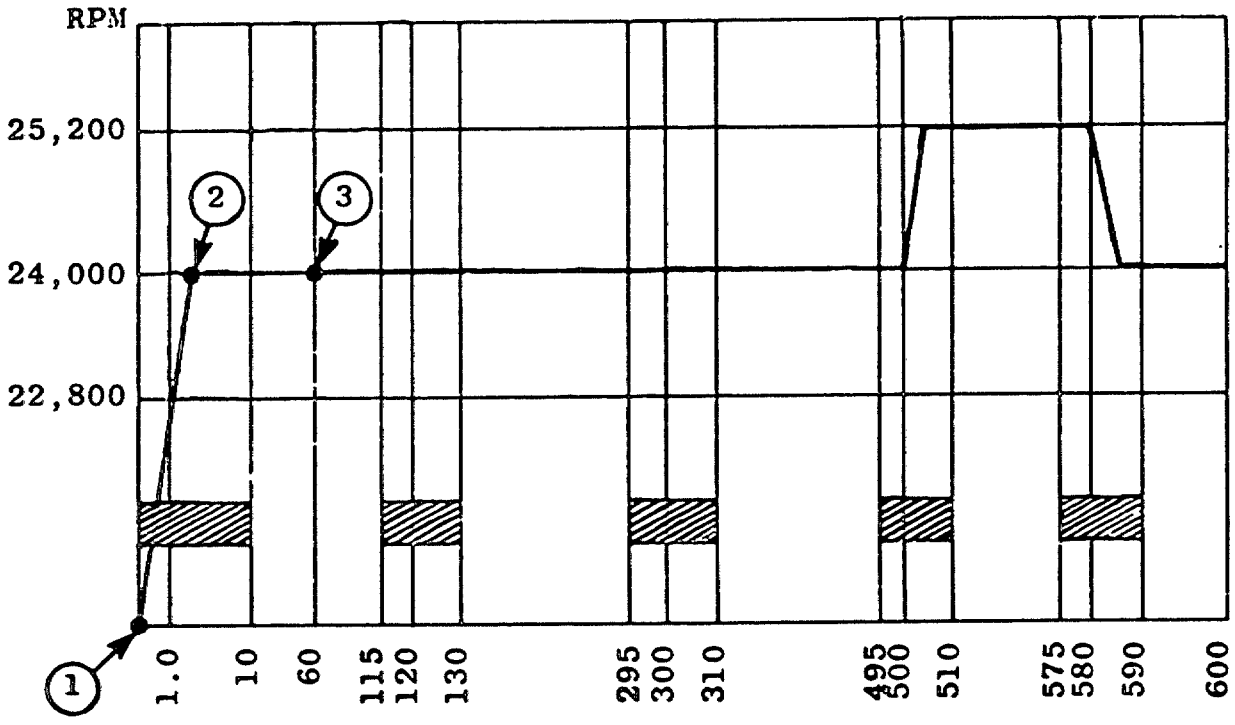


FIGURE 4. AXES OF THE ALTERNATOR-RECTIFIER

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Monitor outputs: ±40 vdc steady state, ±40 vdc pulse, and pulse current

Accelerate from zero speed to 90 percent of 24,000 rpm within 1.0 plus or minus 0.2 seconds.

Rate of change of rpm after pulse load is applied shall not exceed 1000 rpm per second.

1. Steady state load applied as defined in paragraph 3.3.1.3.1 and coolant flow-off.

2. (a) Apply pulse loading at rate of one pulse per 0.5 ±0.1 second. Within 5 seconds after reaching 24,000 rpm, load as defined in paragraph 3.3.1.3.2.

   (b) Coolant flow on for operational temperature tests (4.5.4.1).

3. Coolant flow on for operational life tests (4.5.5).

FIGURE 5. TEST DUTY CYCLE

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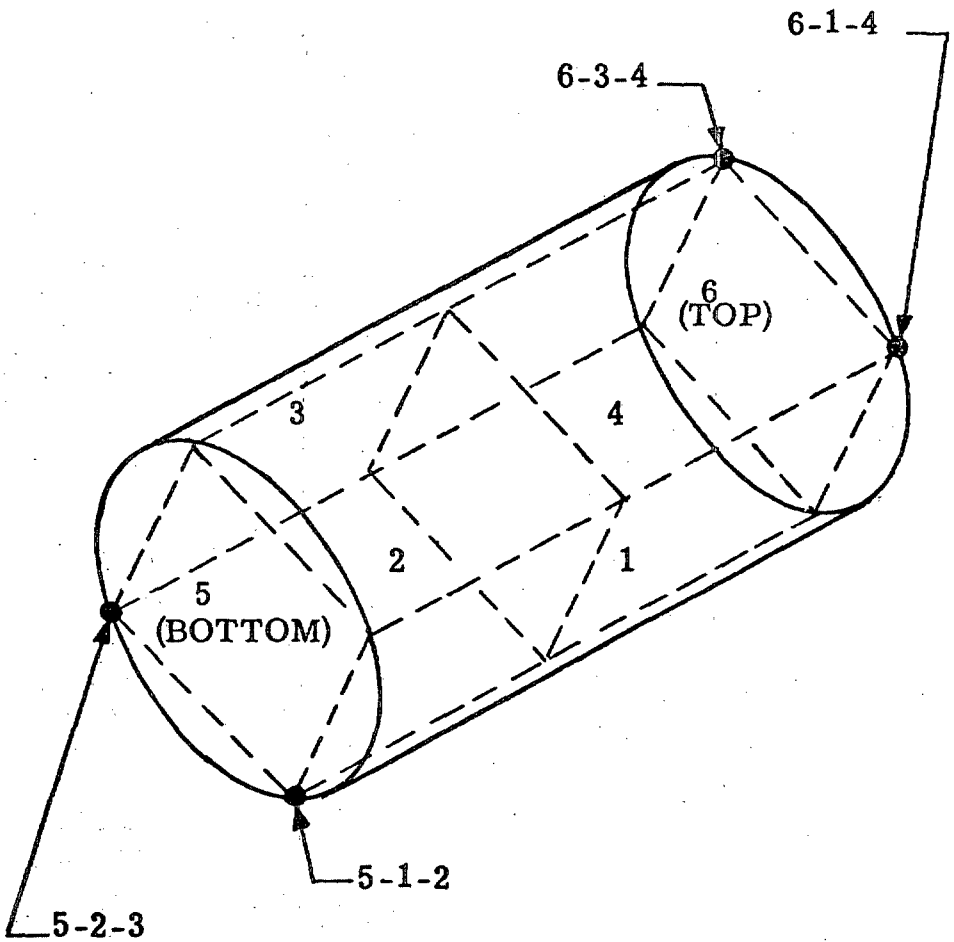


FIGURE 6. DEFINITION OF FACES AND CORNERS OF SHIPPING CONTAINER.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-A-81083C(OS) ALTERNATOR-RECTIFIER (FOR TORPEDO MK 46 MODS 0 and 1)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

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